##
############################

STARTING CONDITION A TEST RUN

Total problems: 15

##
############################

##
############################

Running problem 1/15

##
############################

===
===============

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P01

Problem: Add 1/2 + 1/5

===
===============

$$\frac{1}{2}+\frac{1}{5}=\frac{1\cdot 5}{2\cdot 5}+\frac{1\cdot 2}{5\cdot 2}$$

$$=\frac{5}{10}+\frac{2}{10}$$

$$=\frac{5+2}{10}$$

$$=\frac{7}{10}$$

##

Running problem 2/15

##

==

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P02

Problem: Multiply 3/4 by 2/5

==

$$
\begin{aligned}
\frac{3}{4} \times \frac{2}{5} &= \frac{3 \times 2}{4 \times 5} \\
&= \frac{6}{20} \\
&= \frac{6 \div 2}{20 \div 2} \\
&= \frac{3}{10}
\end{aligned}
$$

##
############################

Running problem 3/15

##
############################

===
===============

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P03

Problem: The fraction 3/4 written as a decimal is

===
===============

$$\frac{3}{4} = 3 \div 4 = 0.75$$

##
############################

Running problem 4/15

##
############################

===
===============

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P04

Problem: If R = {a,b,c,d,e,f}, draw the set that is not a subset of R

===
===============

A set that is not a subset of R would contain at least one element that is not in R. For example, let $S = \{a, b, g\}$. Since 'g' is not in R, S is not a subset of R.

##
############################

Running problem 5/15

##
############################

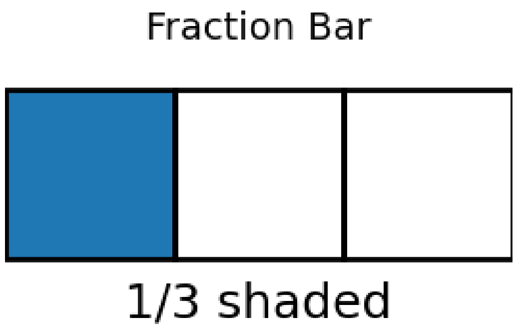
===
===============

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P05

Problem: Draw a rectangular shape that shows the shading of 1/3

==



##

Running problem 6/15

##

==

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P06

Problem: Show that 3 * 7 + 3 * 9 is the same as 7 + 7 + 7 + 9 + 9 + 9

==

$$
\begin{aligned}
3 \times 7 + 3 \times 9 &= (7 + 7 + 7) + (9 + 9 + 9) \\
&= 21 + 27 \\
&= 48
\end{aligned}
$$

##

Running problem 7/15

##

==

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P07

Problem: In a proper fraction, the difference between the numerator and denominator is 5. If the denominator is 7, what is the fraction?

==

$$
\begin{gathered}
\text{Let the fraction be } \frac{n}{d}. \\
\text{Given that the denominator } d = 7. \\
\text{Given that the difference between the numerator and denominator is 5.} \\
d - n = 5 \\
7 - n = 5 \\
n = 7 - 5 \\
n = 2 \\
\text{So, the fraction is } \frac{2}{7}.
\end{gathered}
$$

##
############################

Running problem 8/15

##
############################

===
=================

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P08

Problem: What is 1/8 of 16

===
=================

$$\begin{aligned}
\frac{1}{8} \text{ of } 16 &= \frac{1}{8} \times 16 \\
&= \frac{16}{8} \\
&= 2
\end{aligned}$$

##
############################

Running problem 9/15

##
############################

==

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P09

Problem: The sum of 0.5 and 1.7 is

==

$$0.5 + 1.7 = 2.2$$

##

Running problem 10/15

##

==

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P10

Problem: The top length is a square is 7cm, what is the perimeter?

==

A square has 4 equal sides.

$$
\begin{aligned}
\text{Given side length} &= 7 \text{ cm} \\
\text{Perimeter} &= 4 \times \text{side length} \\
&= 4 \times 7 \text{ cm} \\
&= 28 \text{ cm}
\end{aligned}
$$

##
##############################

Running problem 11/15

##
##############################

==
===================

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P11

Problem: Show the largest to the smallest fraction and why they are so, 1/2, 1/4, 1/3, 1/5

==
===================

To compare fractions, we can find a common denominator.

The least common multiple of 2, 4, 3, and 5 is 60.

$$\frac{1}{2} = \frac{1 \times 30}{2 \times 30} = \frac{30}{60}$$
$$\frac{1}{4} = \frac{1 \times 15}{4 \times 15} = \frac{15}{60}$$
$$\frac{1}{3} = \frac{1 \times 20}{3 \times 20} = \frac{20}{60}$$
$$\frac{1}{5} = \frac{1 \times 12}{5 \times 12} = \frac{12}{60}$$

Now we can compare the numerators:

$30 > 20 > 15 > 12$

So, the fractions from largest to smallest are:

$$\frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \frac{1}{5}$$

##

Running problem 12/15

##

===

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P12

Problem: Draw A U B if A = {1,5,6}, B = {a,c,d}

==

$$A \cup B = \{1, 5, 6\} \cup \{a, c, d\}$$
$$= \{1, 5, 6, a, c, d\}$$

##

Running problem 13/15

##

==

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P13

Problem: Set A = {Even prime numbers greater than 2}, how many members does set A have?

==

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.
The first few prime numbers are 2, 3, 5, 7, 11, ...
An even number is an integer that is divisible by 2.
The only even prime number is 2.
The problem states that Set A contains even prime numbers greater than 2.
Since there are no even prime numbers greater than 2, Set A is an empty set.

$$A = \emptyset$$

The number of members in Set A is 0.

##
###############################

Running problem 14/15

##
###############################

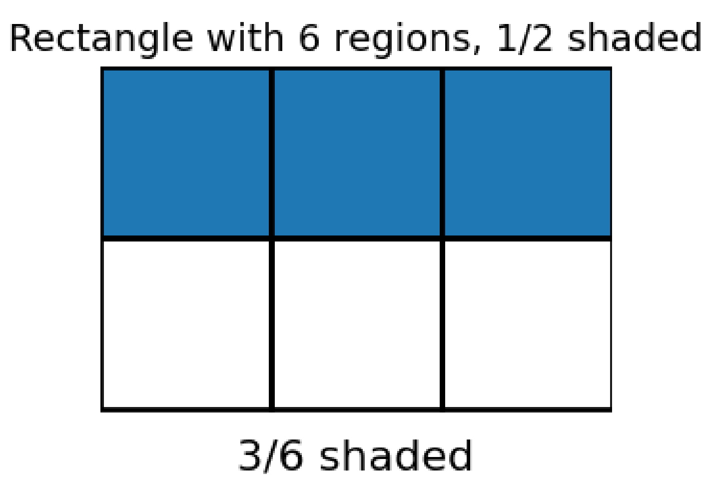
===
===================

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P14

Problem: Draw a rectangle having 6 regions inside, with 1/2 shaded

===
===================



##
###############################

Running problem 15/15

##
############################

===
=================

CONDITION A: Baseline: Direct LLM, no classification, no retrieval, no guidance

Problem ID: P15

Problem: The mixed number 7 2/5 written as an improper fraction is

===
=================

$$
\begin{aligned}
7\frac{2}{5} &= \frac{7 \times 5 + 2}{5} \\
&= \frac{35 + 2}{5} \\
&= \frac{37}{5}
\end{aligned}
$$

##
############################

FINISHED CONDITION A TEST RUN

##
############################